

8.4.36

EE25BTECH11033 - Kevin

Question:

Two sets A and B are as under: $A = \{(a, b) \in \mathbb{R} \times \mathbb{R} : |a - 5| < 1 \text{ and } |b - 5| < 1\}$ $B = \{(a, b) \in \mathbb{R} \times \mathbb{R} : 4(a - 6)^2 + 9(b - 5)^2 \leq 36\}$

1) $A \subset B$

2) $A \cap B$

3) neither $A \subset B$ nor $B \subset A$

4) $B \subset A$

Solution:

Set A:

$$A = \{(a, b) \in \mathbb{R} \times \mathbb{R} : |a - 5| < 1 \text{ and } |b - 5| < 1\} \quad (1)$$

$$|a - 5| < 1 \implies 4 < a < 6 \text{ and } |b - 5| < 1 \implies 4 < b < 6 \quad (2)$$

This describes the interior of a square in the ab -plane. The square is centered at $(5, 5)$ and its sides are the lines $a = 4$, $a = 6$, $b = 4$, and $b = 6$.

Set B:

$$B = \{(a, b) \in \mathbb{R} \times \mathbb{R} : 4(a - 6)^2 + 9(b - 5)^2 \leq 36\} \quad (3)$$

$$\frac{(a - 6)^2}{3^2} + \frac{(b - 5)^2}{2^2} \leq 1 \quad (4)$$

This can be expressed in the form ,

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f \leq 0$$

where,

$$\mathbf{V} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}, \mathbf{u} = \begin{pmatrix} -24 \\ -45 \end{pmatrix} \text{ and } f = 333. \quad (5)$$

This describes the region of an ellipse, including its interior, in the ab -plane. The ellipse is centered at $(6, 5)$ and has a horizontal semi-major axis of length 3 and a vertical semi-minor axis of length 2.

Now, we substitute the maximum possible values for these terms into the expression for the ellipse to see if the condition for Set B is met:

On substitutting $\mathbf{x} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$,

$$\begin{pmatrix} 4 & 6 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix} \begin{pmatrix} 4 \\ 6 \end{pmatrix} + 2 \begin{pmatrix} -24 & -45 \end{pmatrix} \begin{pmatrix} 4 \\ 6 \end{pmatrix} + 333 \quad (6)$$

$$\implies -11 \leq 0 \quad (7)$$

Therefore, every point in Set A is contained within Set B.

This means that A is a subset of B.

Therefore, $A \subset B$

